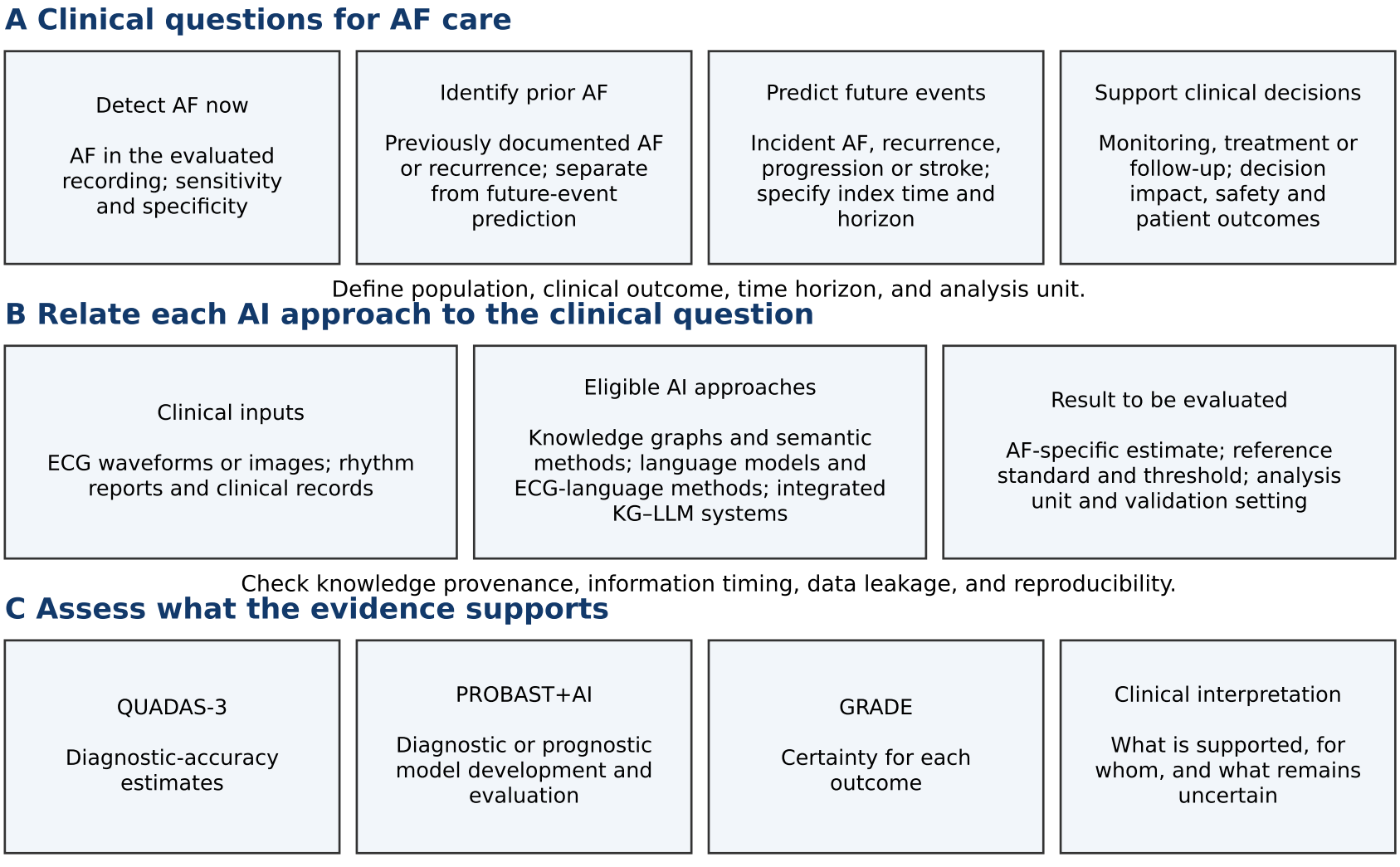
A Clinical questions for AF care
Detect AF now
AF in the evaluated recording; sensitivity and specificity
Identify prior AF
Previously documented AF or recurrence; separate from future-event prediction
Predict future events
Incident AF, recurrence, progression or stroke; specify index time and horizon
Support clinical decisions
Monitoring, treatment or follow-up; decision impact, safety and patient outcomes
Define population, clinical outcome, time horizon, and analysis unit.
B Relate each AI approach to the clinical question
Clinical inputs
ECG waveforms or images; rhythm reports and clinical records
Eligible AI approaches
Knowledge graphs and semantic methods; language models and ECG-language methods; integrated KG–LLM systems
Result to be evaluated
AF-specific estimate; reference standard and threshold; analysis unit and validation setting
Check knowledge provenance, information timing, data leakage, and reproducibility.
C Assess what the evidence supports
QUADAS-3
Diagnostic-accuracy estimates
PROBAST+AI
Diagnostic or prognostic model development and evaluation
GRADE
Certainty for each outcome
Clinical interpretation
What is supported, for whom, and what remains uncertain
Bias and applicability are distinct. Benchmark accuracy does not establish clinical benefit.